Iliya Sabzevari, Ph.D.

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EXPERIENCE

Vice President, *Securities Quantitative Analytics*
***Wells Fargo**, Corporate & Investment Banking* — *June 2024–Present*

- Develop pricing, risk, and analytics to support trading and hedging of xVA, Structured Solutions, and Fixed-Income Structured Notes
- Contribute to a proprietary multi-factor interest rate model used for cross-asset market simulation, forward trade and portfolio pricing, xVA valuation, and PFE, supporting desk risk management and hedging decisions
- Developed forward pricing models for linear and non-linear rates products, supporting trade and risk valuation
- Developed Python-based framework for benchmarking P&L Attribution
- Presented use cases and strategic perspectives on agentic AI software development to the quantitative analytics organization
- Interfaced with cross-functional teams to integrate analytics into production pipelines and decision-making tools
- Work closely with trading to analyze risk profiles, model behavior, and lifecycle risk of xVA and structured products

Senior Consultant, *Quantitative Advisory Services*
***Ernst & Young**, Financial Services & Risk Management* — *September 2022–June 2024*

- Designed quantitative models for derivative pricing, trading, and risk management
- Researched generative AI tools for quantitative research workflows
- Developed cross-asset sensitivity models to optimize hedge ratios and enhance risk allocation in derivative portfolios
- Contributed to a Python-based testing framework to support trade model development and testing

Research Assistant, *Sandeep Sharma Group*
***The University of Colorado at Boulder**, Department of Chemistry* — *Spring 2018–Spring 2022*

- Conducted research in quantum chemistry and quantum Monte Carlo methods
- Developed novel machine-learning inspired algorithms for variance reduction, sampling, and large-scale parameter estimation
- Analyzed large data sets generated by physics-based simulations
- Contributed to the development of Dice, a high-performance C++ modeling package for stochastic and numerical simulations
- Presented interdisciplinary lectures bridging chemistry, physics, mathematics, and computer science

Teaching Assistant
***The University of Colorado at Boulder**, Department of Chemistry* — *Fall 2017–Spring 2022*

- Instructed graduate quantum chemistry, graduate computational chemistry, and undergraduate general chemistry

EDUCATION

The University of Colorado at Boulder — **2017-2022**
***Doctor of Philosophy, Chemical Physics**, 4.0 GPA* — *Boulder, CO*

Advisor: Professor Sandeep Sharma (now at The California Institute of Technology)
Dissertation: *Various Improvements to Quantum Monte Carlo Methods*

The University of Texas at Austin — **2013–2017**
***Bachelor of Science, Chemical Engineering with Honors, Minor in Mathematics**, 3.8 GPA* — *Austin, TX*

SELECTED PUBLICATIONS

3. **Iliya Sabzevari**, Ankit Mahajan, and Sandeep Sharma. An accelerated linear method for optimizing non-linear wavefunctions in variational monte carlo. *The Journal of Chemical Physics*, 152(2):024111, 2020
2. Ankit Mahajan, Nick S. Blunt, **Iliya Sabzevari**, and Sandeep Sharma. Multireference configuration interaction and perturbation theory without reduced density matrices. *The Journal of Chemical Physics*, 151(21):211102, 2019 **"Editor's Choice"**
1. **Iliya Sabzevari** and Sandeep Sharma. Improved speed and scaling in orbital space variational monte carlo. *Journal of Chemical Theory and Computation*, 14(12):6276–6286, 2018

AWARDS AND HONORS

Recognition Award, Quantitative Advisory Services, Ernst & Young — *2022-2024*
Molecular Sciences Software Institute Seed Fellowship — *Fall 2020*
National Science Foundation Graduate Research Fellowship Honorable Mention — *2019*

SKILLS

- Derivatives Pricing & Risk (structured products, exotics, Monte Carlo); Hedging & Sensitivity Analysis (Greeks, scenarios)
- Programming languages: C/C++, Python, Bash, MATLAB, Mathematica, VBA
- Tools and Frameworks: Microsoft Visual Studios, Git, GitHub, LaTeX, Linux/Unix, Windows, Microsoft Office

Hobbies: rock climbing, chess, reading, films, systematic investing